World 4 Bro's: World Bible and Build Plan

Draft 1 for the Antigravity build. Every name, sprite and sound must be original or CC0.

How to use this document

- Each entry has a snake_case ID. Turn every ID into one data definition (a JSON or TypeScript object) so new content never needs engine changes.
- **Does** is the gameplay rule. **FX** is the animation, particles, sound and screen effects.
- Build every object with the same five states: idle, hover, hit, destroy, respawn (see section 9).
- Rarity: C common, U uncommon, R rare. Tags like [night] or [multi] mark special rules.
- Save this file as docs/WORLD_BIBLE.md in the repo so the coding agent can read it.

1. Maps and biomes

The world is one seeded map. Biomes blend using noise and a splat map. Each has a palette, a hazard and a landmark.

- **Bro Meadow (spawn).** Warm greens, gentle hills. No hazard by day. Starter pine, berry bushes, a crooked welcome sign. Teaches chop, craft and campfire. Ambient: butterflies, pollen sparkle, birdsong.
- **Pinewood Thicket.** Dark teal and fog. Easy to get lost, glowing eyes at night. Wood-rich, hollow logs, hanging fireflies. Ambient: light shafts, falling needles, creaking trunks.
- **Squelchy Bog.** Olive and purple. Mud slows you 40%. Bouncy mushrooms, frog choir, gas bubbles that pop, glowing spores at night. Ambient: ripple rings, bubble pops.
- **Rockbottom Ridge.** Grey-blue cliffs. Rockfall after a loud honk. Flint, stone, gold veins, crystals that light the dark. Ambient: echoes, dust motes, skittering pebbles.
- **Sunburn Flats (desert).** Orange with a heat-haze shader. Heat meter and thirst. Cacti, tumbleweeds, a fake oasis mirage that vanishes when you get close. Ambient: dust devils, shimmer.
- **Brrington Tundra.** White-blue. Cold meter and slippery ice. Snowdrifts, a frozen lake for skating, ice fishing holes, snowman watchers. Ambient: gusts, breath puffs, aurora at night.
- **Scrapyard Gulch (gadget biome).** Rust and neon junk. Spring traps and conveyor belts. Source of gadget parts, vending machines, trampolines. Ambient: sparks, flickering signs, distant clanks.

- **The Hollow (nightmare).** Purple-black. Sanity drain and whispers. Dead trees, hanging lanterns, ghost sheets, the best rare loot. Ambient: chromatic wobble, layered whispers, cold fog.
- **Splash Cove (beach and lake).** Turquoise and sand. Deep water drowns you, goose gangs roam. Fishing, crabs, a rubber duck flotilla, a pier. Ambient: foam edges, gulls, waves.
- **Mount Grumbleguts (volcano, endgame).** Black rock, orange glow. Lava and burning ash. Obsidian, ember fruit, the boss arena. Ambient: embers, rumble, pulsing glow.

Special maps and instances

- **Bro Basecamp.** A player-claimed zone for walls, chests, a tent and a firepit ring that creates a safe light zone shared by the party.
- **Echo Caves.** Underground, torch-only. Echoing footsteps, crystals, hidden loot, a gong room.
- **Paper Cutout Plateau.** A pop-up-book area: flat cardboard scenery folds upright as you approach. Home of secrets and the ladder_to_nowhere.
- **Bro-lympics Arena.** Party minigames: egg-and-spoon race, log toss, honk contest. Confetti and a trophy hat.
- **Moon Pond (secret, night only).** Appears under a cheese_moon. Fishing here gives rare cosmetics. Lilies glow and the water mirrors the stars.

2. Natural world objects

- `pine_tree` (C). **Does:** 3 chops, 2 logs, 30% pine cone for replanting. **FX:** shake per hit, wood chips, falls away from you with one bounce, leaves a stump that regrows with a pop-up scale. Thud and leaf burst.
- `oak_tree` (U). **Does:** 5 chops, acorns. A squirrel may drop out and steal one item. **FX:** canopy rustle, acorn rain, squirrel scramble.
- `dead_tree` (C). **Does:** 1 chop, twigs, 10% chance of a ghost sheet. **FX:** creaky sway, dry crack, dust puff.
- `berry_bush` (C). **Does:** pick 3 berries, regrows in a day. **FX:** bush squish, berries pop and arc into the bag, bare twigs until regrowth.
- `grass_tuft` and `twig_sapling` (C). **Does:** harvest for grass and twigs. **FX:** wind sway from shared noise, sparkle on pick, grow back with a tiny pop.
- `wild_carrot` (C). **Does:** pull for food. **FX:** leaves wiggle before the pull, a squeak, flies in an arc.
- `boulder` (C). **Does:** needs a pickaxe, 4 hits, stone and flint. **FX:** sparks, chips, a crack decal that grows each hit, rubble puff, low boom.

- `gold_vein` (U). **Does:** 6 hits, gold nuggets. **FX:** glint sweep every few seconds, coin ping, faint glow at night.
- `crystal_cluster` (U). **Does:** emits light, mine for glow shards (lamps). **FX:** slow hue shift, hum that rises as you approach, shatter into sparkles.
- `glow_mushroom` (U). **Does:** eat for sanity, doubles as a night light. **FX:** pulsing glow, spore trail.
- `buzz_hive` (U). **Does:** honey, but bees chase you unless a fire is near. **FX:** swarm particle cloud, rising hum, smoke calms them.
- `cactus` (C). **Does:** hurts on touch, chop for water pulp. **FX:** wobble, spine particles, water droplets.
- `cattail_reeds` (C). **Does:** fiber for rope. **FX:** sway, fluff puffs as you walk through.
- `fishing_pond` (C). **Does:** fish with a rod. **FX:** ripple rings, bobber plop, tug animation, fish arcs into your hand. Lily pads and frogs that hop away.
- `ice_lake` (U). **Does:** zero-friction skating, cracks under a loud honk. **FX:** scraping skate sound, crack lines, snow spray.
- `lava_vent` (R). **Does:** erupts on a timer after a glow warning. **FX:** ember fountain, heat distortion, ground rumble.
- `campfire` (C). **Does:** light, warmth and cooking. Burns down, needs logs. **FX:** layered flicker, sparks, smoke column, warm bloom, crackle. Steam hiss when it dies.

3. Goofy world objects

- `banana_peel_patch` (C). **Does:** step on it and slide 3 metres. Also a placeable trap that stuns enemies. **FX:** spin with squash and stretch, cartoon whoosh, stars circling the head.
- `springy_stump` (U). **Does:** launches you high to reach ledges. **FX:** squash, boing, camera zooms out in flight, dust ring on landing.
- `whoopee_boulder` (U). **Does:** push it and it lets out a loud pbbbt that scares birds and the lurker for 5 s. **FX:** wobble, green puff, nearby critters freeze or flee.
- `giant_duck_statue` (U). **Does:** squeaks when smacked and hands out a rubber duck. Answers your honk (H) with a honk chorus. **FX:** squash on hit, feather puff, sound ripple rings.
- `old_stuffins` (C). **Does:** a talking scarecrow who gives terrible advice and occasionally a real hint. Swap his hat for a cosmetic reward. **FX:** gibberish speech blips, head wobble, straw puffs.
- `vending_machine_from_nowhere` (R). **Does:** insert 3 gold nuggets for a random goofy item. It may eat the coins and spit out one sock. **FX:** shakes when smacked, blinking neon, clunk drop, a tiny fire if kicked too much.

- `suspicious_bush` (C). **Does:** creeps toward you only while the camera is not facing it, so rotate with Q/E to catch it. Drops berries or a hat when caught. **FX:** eyes blink, leaves rustle, freeze pose when seen.
- `garden_gnome_squad` (U). **Does:** gnomes turn to watch you. Find all ten for a reward (one is golden). **FX:** head-turn tween, hat jiggle, confetti pop on collect.
- `flush_teleporter` (U). **Does:** enter a green outhouse, hear a flush, pop out of another one. **FX:** door slam, swirl particles, spin-out landing, cheeky jingle.
- `disco_stump` (U) [multi]. **Does:** dance (G) near it and every nearby player auto-dances and regains sanity. **FX:** disco ball, glowing floor tiles, beat-synced bounce, light beams.
- `gong_of_regret` (U). **Does:** smack it to ping all teammates on the map, wake sleeping creatures and shake loose rocks. **FX:** shockwave ring, camera shake, long ring-out.
- `fake_campfire` (C) [night]. **Does:** looks like a campfire but gives no warmth. Smack it for confetti and a gotcha sting. **FX:** slightly wrong flicker colour, confetti pop, dramatic horn.
- `sock_tree` (C). **Does:** drops mismatched socks, a craftable warmth item. **FX:** socks flutter down like leaves and wiggle on the branches.
- `leaning_log_tower` (U). **Does:** chop the wrong log and the whole stack rolls over, stunning anyone below. **FX:** wobble warning, rumble, logs bounce and roll with dust, comedic crash.
- `wishing_well_of_lies` (R). **Does:** throw an item in, get a joke fortune and a random item back (50% worse). **FX:** echoey plop, glowing ripples, a hammy narrator voice.
- `mystery_crate_pinata` (U). **Does:** hit it for random loot. **FX:** candy colours, squash on each hit, confetti burst with items arcing out.
- `tumbleweed_buddy` (C). **Does:** rolls past with googly eyes. Catch it for a random item. **FX:** bouncing roll, eyes track you, tiny squeak when caught.
- `ladder_to_nowhere` (R). **Does:** climb it to reveal a tiny hidden platform with a chest. **FX:** the pop-up world folds open at the top, trumpet fanfare.
- `snowman_stalker` (U). **Does:** turns its head to follow you. Look away and it creeps closer, then plops a bucket on your head (vision blocked for 4 s). **FX:** snow crunch, head swivel, clang and muffled screen.
- `clam_of_chance` (U). **Does:** a giant beach clam that snaps. Time your grab for a pearl, mistime for a harmless pinch. **FX:** yawn animation, shine, comic bite with stars.
- `mailbox_with_attitude` (C). **Does:** holds a random letter or hint and bites 1 time in 5. **FX:** flag flip, flap chatter, paper confetti.

4. Goofy gadgets and items

Core tools also needed: axe, pickaxe, shovel, fishing rod, spear, torch, lantern, each with its own swing animation and durability bar.

- `rubber_chicken` (C). **Does:** weapon with low damage and huge knockback. Scares small enemies and never breaks. **FX:** squeak with randomized pitch, stretchy swing, feathers on hit.
- `mega_mallet` (U). **Does:** slow heavy hit that dizzies enemies. **FX:** bonk text, spinning stars over the target, ground crack, hit-stop.
- `boing_boots` (U). **Does:** jump higher and bounce on landing. **FX:** squeak per step, squash and stretch, spring recoil.
- `propeller_beanie` (U). **Does:** glide for 5 s, then a cooldown. **FX:** spinning blades, whirr, leaves swirling below.
- `air_horn` (U). **Does:** upgrade of honk (H). Stuns nearby creatures, wakes sleepers, calls teammates. **FX:** sound-wave rings, camera punch, birds scatter.
- `whoopee_cushion_trap` (C). **Does:** placed trap. The enemy sits, is stunned and embarrassed. **FX:** the sound, red-face particles, sweat drops.
- `confetti_cannon` (U). **Does:** scares night creatures for 6 s and restores sanity. **FX:** muzzle puff, confetti shower, party horn.
- `bean_burrito_booster` (R). **Does:** short dash that costs hunger and leaves a green cloud that repels enemies. **FX:** speed lines, puff trail, comedic rocket sound.
- `umbrella_glider` (U). **Does:** slow fall, blocks rain and the rain_of_frogs. **FX:** open pop, canopy flutter, raindrops bouncing off.
- `bucket_helmet` (C). **Does:** absorbs one hit, then dents. **FX:** clang, dent decal, visible wobble.
- `disguise_kit` (R). **Does:** a fake mustache. Creatures ignore you for 10 s, except the goose. **FX:** smoke puff on equip, twitching mustache.
- `googly_eyes` (C). **Does:** stick on any object. Objects with eyes restore nearby sanity. **FX:** tracking eyes, wobbly pupils, sticker pop.
- `bubble_wand` (U). **Does:** traps a small creature or floats an item. **FX:** iridescent bubble shader, sparkle pop.
- `dancing_lantern` (U). **Does:** a light that bobs to the beat when you dance and restores sanity. **FX:** swaying light, musical notes, colour cycle.

5. Food and cooking with side effects

Cooked in the `bubblebro_cauldron`, whose bubbles change colour by recipe.

- `hiccup_berries`: restores hunger, and your character hiccups and hops slightly for 30 s.
- `giggle_mushroom`: laugh bubbles, wobbly steering, sanity up.
- `chili_of_regret`: big hunger and warmth, then 10 s of fire-breath puffs that scare enemies.
- `mystery_stew`: a random effect, sometimes a tiny voice or a giant head (cosmetic).
- `pancake_stack`: high hunger and sanity. The stack wobbles while carried.
- `rainbow_soup`: your sweat glows in colours and gives a light radius for 60 s.
- `bubble_tea_of_buoyancy`: float on water for 30 s.
- `crispy_sock_chips`: stamina boost, absurd crunch sound.
- `roast_sausage_stick`: simple fire food with a sizzle.

6. Cosmetics

Hats and wearables with their own animation: `traffic_cone_hat` (knocked off by hits), `cheese_hat` (attracts the goose), `flower_crown` (petals when dancing), `tiny_top_hat` (tips when you honk), `pot_helmet` (clangs), `shark_slippers` (flap), `cape_of_dramatic_wind` (flaps with speed). Also face stickers, name colours, emote packs and trail effects (sparkles, bubbles, hearts) so the four bros look distinct.

7. Creatures

- `lurker` [night]: the thing in the title. Invisible beyond the light, glowing eyes at the edge. States: stalk, lunge, retreat. Hates light, fears the whoopee_boulder.
- `honk_goose` (U): steals an item and runs. Honk back to make it flee. Loves the cheese_hat.
- `hop_bun` (C): rabbit-like, flees, drops food. Ears flick.
- `pudge_pigeon` (C): too round to fly well, waddles, drops feathers, steals crumbs.
- `gloomblob` [night]: a puddle-like blob in The Hollow that slides toward you, shrinks in light, drains sanity.
- `pillow_bandit` [night]: hops onto your head and mumbles, draining sanity. Shake it off with Space.
- `waddle_crab` (C): sideways walker on the beach that pinches.
- `dune_noodle` (U): a long wiggly worm that pops out of the sand and trips you. Harmless splash.
- `pocket_sloth` (pet): slow follower that carries a few items and hangs upside down from your pack.

- `quackzilla` (R, boss): a giant rubber duck at the volcano. Squeaky stomps, three phases, drops a golden duck.

8. World events

- `rain_of_frogs`: frogs fall and hop. Umbrellas block them. Frog legs become food.
- `cheese_moon`: yellow moon, cheddar-coloured light, mice appear, the Moon Pond opens.
- `goose_invasion`: waves of honk_geese. Honk to defend and collect loot at the end.
- `traveling_salesbro`: a merchant on a wobbly unicycle who trades odd items for gold.
- `rubber_duck_meteor_shower`: squeaky meteors drop ducks. Craters become ponds.
- `dance_fever`: music starts and everyone must dance or lose sanity. Party reward.
- `fog_horn_morning`: thick fog and a low horn. Visibility is cut, good for ambushes.
- `static_storm` (Scrapyard): gadgets go haywire, conveyors reverse, lights flicker.

9. Effect recipe for every object

Five states: idle (ambient sway or glow), hover (outline plus prompt), hit (shake, 40 to 60 ms hit-stop, particles, sound), destroy (fall, burst, drops that arc to the player), respawn (pop-up scale with Back.easeOut).

Particle presets: wood_chip, leaf_burst, dust_ring, spark, star_bonk, confetti, heart, sweat_drop, fart_cloud (green puff), bubble, feather, snow_puff, ember, spore, coin_glint, ripple_ring, sound_ring.

Screen effects: camera shake (scaled by a reduced-motion setting), zoom punch, chromatic pulse for sanity hits, dusk vignette, slow-mo on boss defeat, lightning flash.

Audio rules: 3 to 5 variants per sound, pitch randomised by about 8%, volume by distance, a rising hum for anything dangerous, a musical sting for rare finds. Layered music for day, dusk, night and danger with crossfades, plus short silly stingers.

10. What else you need

Gameplay systems

- Inventory with a hotbar, stacks, drag-and-drop and equipment slots (hand, head, body, back)
- Crafting tech tree with stations (a brainstormer unlock machine, workbench, cauldron, drying rack)
- Chests and base building: walls, doors, a tent for sleeping, a claim radius

- Seasons (autumn, winter, spring, summer) and weather (rain, snow, fog, storms)
- Farming plots, fishing, hunting, drying food, and spoilage with an icon
- A hint system (the firefly guide), day goals and achievements
- Death rules: ghost mode that can haunt props, a revive item, a respawn penalty, a stats screen
- A boss ladder and difficulty settings
- Map, minimap, teammate pings, an emote wheel and quick chat
- Pets, trading and a friendly-fire toggle

Multiplayer and engineering

- Data registries for items, objects, recipes, biomes, particles, sounds and events
- Seeded deterministic world generation with chunk streaming
- An authoritative realtime server (PartyKit, Colyseus or similar) with room codes. Vercel hosts the site, not the sockets.
- Save and load, autosave, world seeds
- Chat moderation: profanity filter, mute and report, name rules. Chat bubbles reach strangers.
- Reconnect handling and basic anti-cheat. The server validates harvests, crafts and damage.
- A debug panel: spawn any ID, set the time, jump to a biome, toggle night, show colliders and FPS
- Tests (unit tests for the rules, a Playwright smoke test), CI, Vercel deploys, error reporting
- An asset pipeline: glTF with Draco, KTX2 textures, atlas packing, audio sprites
- Performance budgets: instancing, culling, pooled particles, adaptive quality

Art and audio

- A style guide, a palette and one light direction
- A character kit with equipment sockets
- A sound list, music themes, stingers and gibberish voice blips for chat
- A UI kit with icons, fonts and panels
- A licence log in docs/CREDITS.md

Quality and release

- A settings menu: volume, reduced motion, shake, keybinds, graphics quality, language
- Controller and mobile support
- First-10-minutes onboarding and a loading screen

- Accessibility: contrast, colourblind-safe icons, text size
- A privacy notice and terms, with opt-in analytics
- A balance sheet, a playtest plan and a bug tracker
- PWA install support

11. Part 2: Zombie mode overview

Expansion pack. The maps and biomes in section 1 do not change. Roads, points of interest, cars and zombies are new layers on top. The same ID, Does and FX rules apply.

- **Premise:** the bros wake up after a goofy outbreak. Survive, scavenge, fix a car, drive across the biomes and escape. The tone is cartoon slapstick: zombies pop into green puffs with a silly sound. No blood or gore, so it stays playable by everyone.
- **Escape goal:** build a working car and reach the `evac_dock` at Splash Cove. An endless mode stays available for people who only want to survive.
- **Everything from Part 1 stays:** hunger, health, sanity, crafting, goofy objects, the lurker, day and night. Zombies add pressure on top of the old loop.
- **Mode switch:** one config flag (mode: classic or zombie) turns on zombie spawns, high loot, roads and vehicles. Classic mode must keep working.

12. Roads and the road network

Roads are generated from the world seed after the biomes exist. Connect points of interest with a minimum spanning tree plus a few extra loops, route with A* that avoids water and cliffs, smooth into splines, paint them into the terrain splat map, then place props along the edges. Biomes never move.

Road types

- `highway` (Sunburn Flats): fast, wide, long straights, wrecks and roadblocks.
- `country_lane` (Bro Meadow): winding, hedges, farms.
- `logging_track` (Pinewood Thicket): dirt, slow, hides loot.
- `boardwalk_causeway` (Squelchy Bog): planks over mud that creak, with missing planks to patch using `plank_bundle`.
- `switchback_pass` (Rockbottom Ridge): mountain hairpins and a pitch-dark `rock_tunnel`.
- `ice_road` (Brrington Tundra): black ice, low grip, long slides.
- `service_road` (Scrapyard Gulch): conveyors cross the road and shove cars sideways.
- `cursed_road` (The Hollow): fog, cracked asphalt, ghostly headlight reflections.

- `coast_road` and `pier` (Splash Cove): tide splashes, sand drifts.
- `lava_bridge` (Mount Grumbleguts): cooled lava spans with glowing cracks.

Road pieces and props

`junction`, `roundabout_of_doom` (zombies circle it), `toll_booth` (loot, and the barrier arm bonks you), `railway_crossing` (train event), `bridge` (a choke point that wrecks can block), `tunnel`, `road_sign` (goofy directions that sometimes lie), `guardrail`, `traffic_cone` (also a zombie hat and a noise maker), `road_barrier`, `billboard_ad` (original fake brands), `parked_wreck` (loot and part source).

Road rules

- **Does:** roads give on-foot players +10% speed and full grip to cars. Off-road costs grip and fuel.
- **Hazards:** oil slicks, potholes, spike strips, fallen trees (clear with an axe), roadblocks (clear with a tow, a crowbar, or by driving through at the cost of car health).
- **Risk:** zombies cluster at wrecks and junctions, so roads are faster but riskier.
- **FX:** tyre-mark decals, dust trails on dirt, sparks on guardrail scrapes, puddle splashes in rain, headlight cones in fog.

13. Points of interest along the roads

POIs sit beside roads inside the existing biomes. Each has a loot table, a zombie density and one gimmick.

- `gas_station` (highway, lanes): fuel, snacks, tyre kits. Medium zombies. Touching a pump trips an alarm that pulls a mini horde.
- `motel_no_vacancy` (desert, coast): beds for safe sleep, vending machines, a key rack with car keys. Low zombies, surprises behind doors.
- `auto_garage` (Scrapyard, lanes): the best source of car parts and tools, plus a `car_lift` that speeds up repairs. Medium zombies.
- `police_station` (Meadow edge): flares, roadblock gear, locked cells with loot. High zombies. Tripping the alarm sets off a siren.
- `farmhouse` (Meadow, Thicket): food, jerrycans, tools, a tractor. Low zombies, barn hordes.
- `supermarket` (Meadow): lots of food and shopping carts. High zombies, aisle ambushes.
- `hospital_ruins` (Hollow edge): antidote gel and bandages. High zombies and sanity drain.
- `logging_camp` (Thicket): axes, planks, a loud chainsaw.

- `radio_tower` (Ridge): the rescue radio quest, signal flares. Climb it to reveal the map.
- `drive_in_cinema` (Flats): the film sound lures zombies, loot sits in the booths.
- `ice_cream_depot` (Tundra): an ice cream truck wreck and frozen loot.
- `checkpoint_barricade` (Tundra, Ridge): cartoon crates with high loot and high risk.
- `lighthouse_diner` (Cove): food and the boat key.
- `volcano_scrap_vault` (Grumbleguts): jackpot loot guarded by a boss.
- `evac_dock` (Cove): the escape point. Arriving starts a final defend-the-engine event.

14. Dumb cars: the vehicle roster

All cars are goofy, arcade-handled and original (no real brands). Each has seats, trunk slots, a top speed, a fuel use, a quirk and a list of required parts.

- `rusty_hatchback` (C): 4 seats, 8 trunk slots, average everything. **Quirk:** backfires a puff that startles zombies. Honk is a sad trombone.
- `clown_car` (U): 6 seats, 4 slots. **Quirk:** more passengers than seats appear, a bouncy ride, a bike-horn honk, and zombies it bumps burst into confetti. Needs a `big_red_nose` as its key.
- `ice_cream_truck` (U): 3 seats, 20 slots, slow. **Quirk:** its looping jingle pulls zombies in like a magnet, so you can lure hordes on purpose. The freezer stops food spoiling.
- `golf_cart_deluxe` (C): 2 seats, very quiet, low fuel use. **Quirk:** crawls up hills at a hilarious speed. Great for sneaking.
- `bathtub_buggy` (R): 1 seat, floats like a boat. **Quirk:** leaves a bubble trail and a soap slick.
- `lawn_tractor_racer` (C): 1 seat, tows a trailer, strong off-road grip. **Quirk:** lower the blades to clear grass and light zombies with cartoon puffs.
- `school_bus_of_chaos` (R): 12 seats, 30 slots, a rolling base. Slow turning. **Quirk:** the stop-sign arm swings out and bonks zombies.
- `hot_dog_van` (U): 3 seats, 16 slots, a griddle for cooking on the move. **Quirk:** the sizzle and smell cloud attract zombies and the goose.
- `shopping_cart_runner` (C): 1 seat, no engine, no fuel. Coast downhill. **Quirk:** a wobbly wheel that fights every turn.
- `pizza_scooter` (C): 2 seats, fast and fragile, a 6-slot delivery box. **Quirk:** pops a wheelie when you boost.
- `monster_truck_of_questionable_taste` (R): 2 seats, runs over wrecks and zombies, loud, thirsty. **Quirk:** googly eyes on the grille and a roar for a horn.
- `banana_van` (U): 4 seats. **Quirk:** drops banana peels behind it that slip zombies.

Fuel: `gas_can`, siphoned fuel (with a `hose`), and goofy emergency fuels listed under jank fixes. Fuel shows on the HUD, and running dry rolls the car to a stop with a cough.

15. Car repair: find and fix the parts

A car is a list of parts. Each part has a condition (0 to 100) and a repair rule. A car runs only when every required part is above 20%.

Required parts: `battery`, `spark_plugs`, `fuel`, `engine_block`, `ignition_key` (or hotwire with a mini-game), 4 `tires`, `radiator_coolant`, `fan_belt`.

Optional parts: `headlights`, `horn`, `windshield`, `doors`, `bumper`, `roof_rack`, `radio`, `trunk_lock`, `seat_covers` (sanity bonus), `fuzzy_dice` (luck).

How repair works (Does)

1. **Find a car.** Parked wrecks, garages, barns, roadsides. Each spawns with 30 to 60% of its parts missing, seeded so runs are repeatable.
2. **Inspect.** Hold on the car to open the repair panel. Every part is a card: missing, broken, damaged or fine. Missing parts appear as glowing ghost outlines in the engine bay.
3. **Find parts.** Garages, wrecks and shops. Parts come in junk, standard and shiny quality.
4. **Fit a part.** A hold-to-repair ring. Tools shorten it: `wrench`, `ratchet`, `jack`, and the garage `car_lift`.
5. **Fix damaged parts.** `duct_tape` (a patch that caps at 40%), `welding_kit`, `tire_patch`, `jump_leads` for the battery.
6. **Start the engine.** A short crank mini-game: hold, then release in the green zone. A failure floods the engine and costs time. Starting is loud and pulls zombies in.
7. **Fuel up.** Siphon or pour from a `gas_can`.
8. **Driving wears parts.** Zombie crowds dent the bumper, rough roads rattle the belt, hits crack the windshield. Smoke at 30% engine, fire at 10%. Put it out with a `fire_extinguisher` or lose the part.

Co-op: up to four bros repair in parallel, each on a different part, with name tags over the engine bay. Roles are free: driver, mechanic, lookout, gunner. A `mechanic_hat` makes repairs 20% faster.

Jank fixes (the funny part). With no proper part you can use junk. It works, but the car gains a quirk shown on its card.

- `bucket_headlight`: a narrow cone of light that swings loosely.
- `rubber_chicken_horn`: honks as a squeak and scares small enemies.
- `pancake_stack_tire`: a bouncy ride that boings over bumps, at lower top speed.

- `sock_fan_belt`: works at 50% and puffs sock-scented steam.
- `bean_burrito_fuel`: a speed boost that leaves green clouds and burns your food.
- `bucket_of_water_radiator`: works until the water runs out.
- `googly_eyes_dashboard`: sanity bonus for riders.
- `cheese_wheel_tire`: rolls oddly and attracts the goose.

FX: sparks, wrench and ratchet clicks, steam hiss, a ding when a part clicks in, the car shaking while it is fixed, a coughing engine and finally a triumphant roar with a camera zoom-out, a honk and a sparkle burst.

16. Driving model

- **Arcade physics, not a simulator:** acceleration, brake, handbrake drift. Grip per surface: asphalt 1.0, dirt 0.7, grass 0.65, sand 0.6, snow 0.45, mud 0.4, ice 0.2.
- **Camera:** follows behind and pulls out with speed. Q and E still rotate it, with a lock-behind option.
- **Collisions:** the car pushes light props, bounces off trees and boulders with part damage, and plows through zombies with a slowdown based on how many it hits.
- **Noise:** engine noise radius grows with speed and car type. The golf cart is quiet, the ice cream truck is loud.
- **Controls:** WASD, Space handbrake, H honk, F enter and exit. On mobile, a steering thumbstick, two pedals and a honk button.
- **Passengers:** seat swap, trunk access, and a gunner seat for window attacks.
- **Parking:** park at Basecamp to protect parts. A covered garage prevents weather damage.
- **FX:** wheel-spin dust, drift smoke, motion blur at speed, headlight cones in fog, engine hum pitched by speed, tyre squeal, suspension bounce.

17. Zombies

Zombies live in the existing biomes with tinted variants. All use the cartoon pop on defeat.

- `shambler` (C): slow, groans, the standard.
- `jogger` (U): fast, comes in packs, appears at dusk.
- `traffic_cone_zombie` (C): the cone blocks the first hit. Knock it off and the zombie topples.
- `clown_zombie` (U): squeaky honks that attract others.
- `bloater_balloon` (U): inflates and floats, then pops into a slowing cloud.

- `brute_in_a_bathrobe` (R): big and slow, stuns, headbutts cars.
- `screamer_karaoke` (R): holds a mic. Its scream calls a horde. Flares are the answer.
- `crawler_roomba` (C): low, hides in grass, trips you.
- `zombie_goose` (U): honks, bites, steals items.
- **Biome variants:** `frost_zombie` (Tundra), `husk_zombie` (Desert), `mud_zombie` (Bog), `ember_zombie` (Volcano), `ghost_sheet_zombie` (Hollow).
- `highway_hulk` (R, boss): a giant zombie stitched from wrecked cars, guarding the road to the evac dock. Drops a rare engine part.

Zombie systems

- **Senses:** vision cone, hearing from noise, smell for food, attraction to light at night.
- **Noise events:** footsteps (sprinting is louder), engines, horns, loud tools, gongs, alarms. Show the noise radius as a ring in debug mode.
- **AI states:** wander, investigate, chase, attack, stunned, flee fire, follow horde.
- **Hordes:** size scales with the day and with noise. Every 7th night is a `horde_night` with a siren warning at dusk.
- **Infection:** bites add an infection meter. Cure it with `antidote_gel`. At 100% you turn into a `zombie_bro` that teammates can cure with a revive item.
- **Performance:** crowd instancing, a vertex-animation shader, LOD swaps, flow-field pathfinding on a grid, far zombies updating at 2 Hz, pooled spawns, and a separate active-zombie budget for phone and desktop.

Goofy objects meet zombies

- `banana_peel_patch`: zombies slip and slide.
- `disco_stump`: nearby zombies are forced to dance for 6 s.
- `gong_of_regret`: calls every zombie in a huge radius. A dangerous dinner bell.
- `whoopee_boulder`, `rubber_chicken` and `giant_duck_statue`: noise decoys.
- `fake_campfire`: pulls zombies in instead of warming you.
- `confetti_cannon`: stuns a crowd.

18. High loot system

The goal is that you find things all the time, so the fun is choosing what to carry. Danger and weight keep the economy fair.

- **Tiers:** Junk (grey), Common (white), Useful (green), Rare (blue), Jackpot (gold).

- **Containers:** `glove_box`, `car_trunk`, `suitcase`, `locker`, `toolbox`, `fridge`, `medicine_cabinet`, `cash_register`, `piggy_bank` (a mini jackpot), `supply_drop_crate`, plus the existing vending machine and pinata.
- **Loot density setting:** low, normal, high. High is the default: about 2x containers and +1 item per container.
- **Weight and slots:** 12 backpack slots to start, up to 24 with bags. Carrying over 100% weight slows you. Cars add trunk slots and bases add chests.
- **Risk and reward:** opening a Jackpot crate fires a light beam and a siren sting that attracts zombies for 20 s. Locked containers need a `crowbar` or `lockpick`, which makes noise.
- **Hot zones:** police_station, checkpoint_barricade, hospital_ruins and volcano_scrap_vault have 3x loot and dense zombies.
- **Restock:** containers refill every 3 days, only when no player is within 60 m. Wrecks refill parts more slowly.
- **Loot tables:** one weighted table per container and POI in data (example: garage is 50% parts, 25% tools, 15% junk, 10% rare).
- **Guarantees:** within 100 m of spawn there is always a campfire kit, a weapon and a snack. Car parts bias toward roads so the car quest stays winnable.
- **FX:** sparkle by tier, a pop sound with rising pitch, items arc into the loot panel, a confetti burst on Jackpot, container lid animations.
- **Multiplayer:** a setting for personal loot rolls (no arguments) or shared loot.

19. Weapons and defenses

Mostly melee and goofy, readable and fun. Everything has durability.

- **Melee:** `crowbar`, `nail_bat`, `frying_pan`, `traffic_sign_axe`, `pool_noodle_of_justice` (long reach, tiny damage, big knockback), plus the existing `rubber_chicken` and `mega_mallet`.
- **Ranged and throwables:** `slingshot`, `water_balloon` (slows), `road_flare` (lures or scares), `firework_launcher` (loud, area damage), `foam_dart_blaster`, `cream_pie` (blinds).
- **Defenses:** `spike_strip`, `barricade_planks`, `sandbag`, `searchlight_tower`, `clam_trap`.

20. New systems checklist for this expansion

- A mode and rules file: classic or zombie, loot density, zombie density, horde frequency, infection on or off, friendly fire, evac goal on or off.

- A seeded road generator with a debug overlay, and a POI placer with spacing rules and a reachability check. Every required part and the evac dock must be reachable.
- A vehicle entity (server-owned in multiplayer) with seats, parts, repair panel UI, fuel, damage, noise and saves.
- A zombie crowd system with instancing, flow fields and noise propagation, plus a performance budget.
- A loot system with tables, weights, tiers, restock and guarantees.
- A quest chain: the `rescue_radio` leads to parts, keys and the evac dock. The firefly guide gives hints.
- Balance: the parts for one car should be findable within about 3 days at normal density. Keep a playtest sheet for fuel, parts, zombie damage and loot rates.
- UI: repair panel, fuel and speed HUD, noise meter, infection meter, weight bar, minimap with roads and POI icons.
- Audio: engine sounds per car, horns, layered zombie groans, a horde siren, and a car radio with original goofy stations.
- Save data: part conditions, POI looted state, road seed.
- Content safety: cartoon tone by default, a gore-free setting, no real car brands, and a licence log for every asset.

21. Part 3: Master item catalog

Part 3 adds items, loot, settlements, shelters and the map-generation plan. Numbers are starting values to tune. D = damage, S = speed, N = noise from 0 (silent) to 5 (alarm), Dur = uses before it breaks. Tiers match section 18: J junk, C common, U useful, R rare, X jackpot.

Melee weapons

- `stick` (J): D 5, S fast, N 1, Dur 25. The starter. Snaps with a cartoon crack.
- `stale_baguette` (J): D 6, S fast, N 0, Dur 20. Breaks into crumbs that pigeons chase.
- `nail_bat` (C): D 14, S medium, N 2, Dur 60. Nails glint, thwack sound.
- `crowbar` (C): D 12, S medium, N 2, Dur 80. Also opens locked containers and clears roadblocks.
- `frying_pan` (C): D 10, S medium, N 3, Dur 70. A bong sound and a 15% stun chance.
- `hockey_stick` (C): D 11, S medium, N 2, Dur 60. Long reach.
- `pool_noodle_of_justice` (C): D 2, S fast, N 0, Dur 999. Double reach, huge knockback, a boing on hit.
- `traffic_sign_axe` (U): D 20, S slow, N 3, Dur 50. The sign shows a random goofy symbol.

- `fire_axe` (R): D 28, S medium, N 3, Dur 80. Chops doors, trees and zombies.
- `chainsaw_of_regret` (R): D 35 over time, N 5, burns fuel. Wobbles in your hands, zombies pop in green puffs.
- Existing: `rubber_chicken`, `mega_mallet`.

Ranged weapons

- `slingshot` (C): D 8, N 1, ammo pebbles.
- `bow_of_branches` (C): D 12, N 1, slow draw, arrows can be picked up again.
- `foam_dart_blaster` (U): D 6, rapid fire, N 1, ammo foam_darts (recoverable).
- `crossbow_of_corks` (U): D 18, N 1, slow reload, ammo corks (reusable).
- `squirt_cannon` (U): D 0, slows and soaks, puts out fires, refills at ponds.
- `plunger_launcher` (U): sticks a zombie to a wall for 4 s.
- `potato_cannon` (U): D 15, N 3, ammo potatoes (which are also food).
- `flare_gun` (U): D 5 plus burn, lights an area, lures or scares.
- `nail_gun_of_justice` (R): D 10, rapid fire, N 3, ammo nails.
- `firework_launcher` (R): D 40 in an area, N 5, ammo bottle_rockets.
- Existing: `confetti_cannon`, `bubble_wand`.

Throwables

`water_balloon` (J, slows), `cream_pie` (C, blinds 4 s), `road_flare` (C, light, lures or scares), `marbles` (C, scatter and slip), `banana_peel` (C, slip), `alarm_clock_decoy` (C, rings after 5 s and pulls zombies), `stink_sock_bomb` (U, zombies stagger away), `smoke_pot` (U, cover), `boombox_decoy` (U, plays music and pulls a horde), `bowling_ball` (U, heavy, strikes zombies like pins), `glitter_bomb` (R, blinds and marks a crowd).

Traps and defenses

`tripwire_cans` (C, noise alert), `spike_strip` (C), `barricade_planks` (C), `sandbag` (C), `shopping_cart_barrier` (C, blocks roads and doors), `whoopee_cushion_trap` (C), `barbed_wire_tangle` (U), `sprinkler_slip_n_slide` (U, soaks and slides zombies), `clam_trap` (U), `scarecrow_decoy` (U), `searchlight_tower` (R), `foam_turret` (R, auto-fires darts, needs power), `flame_pit` (R).

Armor and wearables

- `bucket_helmet` (C): absorbs 1 hit, then dents.
- `bike_helmet` (C): less crash damage.
- `pot_helmet` (C): +10% head protection, clangs.
- `pillow_vest` (C): -20% bite chance, feathers burst out when hit.
- `cardboard_armor` (J): light protection, weakens when soaked by rain.

- `hockey_pads` (U): -20% melee damage taken.
- `rain_poncho` (C), `winter_coat` (U), `sun_hat` (C), `snow_boots` (C), `swim_floaties` (C, float): weather gear tied to the biome meters.
- `hi_vis_vest` (C): teammates show brighter on your minimap.
- `fanny_pack_of_holding` (R): +4 quick slots.
- Backpacks: `school_backpack` (+6 slots), `hiker_backpack` (+12 slots).

Medical and survival

`bandage` (C), `painkillers` (C), `first_aid_kit` (U), `antidote_gel` (R, cures infection), `energy_drink` (C, stamina then jitters), `thermos_of_cocoa` (U), `water_bottle` (C), `water_purifier` (U), `hand_warmers` (C), `sunscreen` (C), `gas_mask` (R, blocks bloater clouds), `sleeping_bag` (C), `tent_kit` (C).

Tools and utility

`axe`, `pickaxe`, `shovel`, `wrench`, `ratchet`, `jack`, `welding_kit`, `lockpick` (slow, noisy), `bolt_cutters` (U, cut fences and padlocks), `flashlight` (C, cone, uses batteries), `headlamp` (U), `walkie_talkie` (U, teammate range), `binoculars` (U), `map_scrap` (C, reveals a region), `compass` (C), `rope`, `duct_tape`, `hose`, `jerrycan`, `fishing_rod`, `multi_tool` (R), `grappling_hook_of_doubt` (R, reaches rooftops, sometimes misses).

Movement toys

`skateboard` (U, fast and silent on roads), `roller_skates` (U), `pogo_stick` (U, bounces over fences), `zipline_kit` (R).

Vehicle items (see section 15)

`battery` (U), `spark_plugs` (C), `engine_block` (R), `fan_belt` (C), `radiator` (U), `tire` (C), `fuel_pump` (U), `muffler` (U, less engine noise), `bull_bar` (U, plows zombies), `armor_plating` (R), `spoiler_of_speed` (U, +5% top speed), `turbo_can` (R), `horn_pack` (C, swaps your horn: trombone, bike horn, duck, rooster), `air_freshener_tree` (C, small sanity boost), `paint_can` (C, colours), `funny_license_plate` (C, cosmetic).

Crafting materials

`wood_log`, `plank`, `stone`, `flint`, `scrap_metal`, `nails`, `cloth_scrap`, `rope`, `glue`, `wire`, `spring` (from mattresses), `gears`, `empty_bottle`, `battery_cell`, and `gunk_goo` (dropped by zombies, used for jank fuel and slip traps).

Junk and goofy loot

`mismatched_sock`, `rubber_duck`, `whoopee_cushion`, `fake_mustache`, `lava_lamp` (light plus a calming effect), `garden_gnome`, `trophy_cup`, `broken_toaster` (sells or scraps for parts), `novelty_mug`, `inflatable_flamingo` (floats, works as a decoy), `kazoo` (decoy),

`disco_ball_shard`, `giant_novelty_check` (X), `golden_toilet_seat` (X).

Currency and ammo

- `shiny_buttons`: the common currency for traders, found everywhere in small amounts.
- Ammo: `pebbles`, `foam_darts`, `corks`, `nails`, `potatoes`, `bottle_rockets`, `flare_cartridges`, `batteries_aa`.

Keys and quest items

`house_key`, `car_key`, `keycard_red`, `keycard_blue`, `keycard_green`, `padlock_code_note`, `rescue_radio_part_1`, `rescue_radio_part_2`, `rescue_radio_part_3`, `boat_key`, `map_fragment` (points to a treasure spot), `signal_flare`, `evac_beacon`.

22. Loot tables by place

Each table lists category weights out of 100. Every building rolls one theme, and each room type shifts the weights.

- `house_kitchen`: food 40, drinks 15, tools 10, junk 25, medical 5, rare 5.
- `house_bedroom`: wearables 35, junk 30, medical 15, melee 10, rare 10.
- `house_garage`: car parts 30, tools 30, fuel 15, melee 10, junk 10, rare 5.
- `shop_general`: food 30, drinks 15, junk 25, tools 15, medical 10, rare 5.
- `pharmacy`: medical 60, drinks 10, junk 15, rare 15 (antidote gel lives here).
- `diner`: food 55, drinks 25, junk 15, rare 5.
- `school`: junk 40, sports gear 20 (bats, hockey sticks, bowling balls), throwables 15, medical 10, keys 5, rare 10.
- `office`: junk 40, batteries 15, tools 15, maps 10, keycards 10, rare 10.
- `police_locker`: melee 20, throwables 20, armor 20, keys and keycards 15, ammo 15, rare 10.
- `farm_barn`: tools 30, food 25, fuel 20, melee 15, vehicle parts 10.
- `workshop`: car parts 50, tools 30, fuel 10, rare 10.
- `outpost_crate`: ammo 25, armor 20, medical 15, ranged 15, keycards 10, rare 15.
- `vault_jackpot`: rare 50, jackpot 30, keycards 10, medical 10.
- `zombie_drop`: gunk_goo 40, shiny_buttons 30, junk 20, rare 10.

23. Balance rules for items and weapons

- **Trade-offs:** every weapon gives up something: damage, noise, weight or durability. Nothing is best at everything.
- **Noise** uses the same 0 to 5 scale as the zombie noise system, so weapons feed the stealth loop directly.
- **Weight scale:** light 0.2, medium 1, heavy 3, very heavy 8. Weight and slots are both limits.
- **Weapon mods:** a `duct_tape` grip (+durability), `nails` (+damage, more noise), `googly_eyes` (sanity), a `bell` (louder, but lets you pull zombies away from teammates).
- **Progression tiers:** 0 hands and sticks, 1 scavenged melee, 2 ranged toys and traps, 3 gadgets and turrets, 4 armored vehicles.
- **Scarcity:** the first night is guaranteed one melee weapon. Ammo is common but light, so you carry it deliberately.
- **Repairs:** melee weapons can be repaired at a `workbench` with scrap_metal and glue, to a lower max each time.

24. Settlements: towns, villages and small places

Settlements sit on top of the existing biomes. Nothing in section 1 moves. Each has a size, a biome fit, a zombie density, a loot level and a signature gimmick.

- `town` (any flat biome, 25 to 60 buildings): grid streets, town hall, school, clinic, supermarket, police station, gas station, apartments, a park and a water tower. Highest loot, highest zombies. Gimmick: a siren on the town hall that, once rung, draws every zombie in the valley.
- `village` (Meadow, Thicket, Tundra, 8 to 20 buildings): a main street, a well and a market square. Medium loot. Gimmick: a bell that wakes zombies but also calls helpful survivors.
- `hamlet` (any, 3 to 6 buildings): a few houses along a lane. Low zombies, good for the early game.
- `farmstead` (Meadow, Flats, 3 to 5): farmhouse, barn, silo, tractor. Food and fuel.
- `strip_stop` (highways, 3 to 5): gas station, diner and motel together. The roadside lifeline.
- `harbor_town` (Cove): pier, fish market, boats, lighthouse. Boats and the evac route.
- `stilt_village` (Bog): houses on stilts joined by boardwalks. Mud below, so zombies crawl.

- `desert_trading_post` (Flats): adobe shops, a water tower, the drive-in. Water is the loot.
- `ski_lodge_village` (Tundra): a lodge, chairlift and cabins. Warm shelter from the cold meter.
- `mining_camp` (Ridge): shafts, ore carts, a bunkhouse. Dark tunnels with tool loot.
- `logging_camp` (Thicket): sawmill, log stacks, a cookhouse.
- `junkyard_town` (Scrapyard): bus houses and container homes. Parts everywhere.
- `ghost_village` (Hollow): half-swallowed by fog, with sanity drain and rare loot.
- `volcano_observatory` (Grumbleguts): a research station with a jackpot vault.

25. Lost outposts and hidden places

- `ranger_tower` (Thicket, Ridge): climb it to reveal the map. Has a supply stash.
- `weather_station` (Tundra): radio parts and storm warnings.
- `survey_camp` (Flats): maps and binoculars.
- `abandoned_bunker` (Ridge): keycard doors, pitch dark, jackpot loot.
- `container_fort` (Scrapyard, Cove): stacked shipping containers that can become a base.
- `lighthouse_outpost` (Cove): with a generator its beam sweeps and clears zombies in a cone at night.
- `pancake_cult_compound` (Hollow): a goofy cult of pancake worshippers with silly relics.
- `abandoned_theme_park` (Meadow edge): a haunted carousel, prize booths full of loot and a goofy mid-boss.
- `circus_camp` (Flats): clown_zombies and a big top full of props.
- `crashed_cargo_plane` (any): looting the supply pile is very loud, and a ring of zombies arrives.
- `hermit_cabin` (Thicket): a hermit trader with odd tastes.
- `sunken_bus` (Bog): loot underwater, needs the bubble_tea_of_buoyancy.
- `treasure_x_spots`: found with a `map_fragment`, dig with a shovel.
- `bro_museum`: an easter-egg hall showing the four bros as statues.

26. Building kit and interiors

Building types: `house_small`, `house_two_story`, `cabin`, `bungalow`, `apartment_block`, `shop`, `supermarket`, `pharmacy`, `diner`, `garage`, `gas_station`, `school`, `chapel`, `town_hall`, `police_station`, `clinic`, `warehouse`, `barn`, `silo`, `water_tower`, `motel`, `shed`, `playground`.

Interior rules

- Buildings use grid footprints (about 6x6 m up to 16x12 m) with 1 to 3 floors.
- Rooms come from templates (kitchen, bedroom, office, storage), and each picks a loot table from section 22.
- Doors are unlocked, locked, boarded or barricaded. Windows can be climbed through, which makes noise.
- The pop-up look: roofs and walls fade and fold away as you enter, so you can see inside without the camera clipping.
- Interiors generate lazily, only when a player is within about 40 m, using a hash of the seed and the building ID so they are always identical.
- Light: power can be on or off. A flashlight cone reveals dark rooms. Flickering bulbs hint at zombies nearby.
- Hiding spots: `wardrobe`, `under_bed` and `dumpster`, with funny cramped animations. Zombies sniff and wander off if you stay quiet.

Settlement conditions (set per settlement): pristine, worn, burnt, flooded, collapsed, overgrown, barricaded by survivors. Each changes props, light, loot and zombie behaviour.

27. How the map generates

1. **Seed and biomes:** the existing terrain and biome generation runs first and is never changed.
2. **Suitability map:** score every cell for settlements using flatness, distance to water, distance to other settlements and a biome weight.
3. **Site picking:** Poisson-disc sampling with per-type minimum distances picks sites. Starting targets for a normal map: 2 towns, 6 to 8 villages, 8 to 12 hamlets and farmsteads, 6 to 10 outposts, 10 to 12 strip stops. Keep towns far apart (400 m or more).
4. **Site types:** the biome and score decide each site's type from section 24.
5. **Roads:** connect every site with the road graph from section 12 (minimum spanning tree plus loops, pathfinding around water and cliffs, splines). Each settlement's main street aligns with the road passing through it.

6. **Layout algorithm by type:** grid (towns), strip along the road (stops), radial around a well or plaza (villages), cluster (farmsteads), shoreline (harbor), stilts and boardwalks (bog), ring fence (outposts).
7. **Lots:** split blocks into lots with binary space partition, assign a building type by zone (shops at the center, houses toward the edge, farms and warehouses at the outskirts), then check clearance so nothing overlaps.
8. **Fill:** props, fences, wrecks, street lamps, containers with loot and zombie spawn points.
9. **Condition:** pick the ruin level by distance from spawn and from the evac dock, so danger grows as you head toward the goal.
10. **Shelters:** mark claimable safe houses (section 28) in every settlement and outpost.
11. **Validate:** the three radio parts, the keys, a car, fuel and the evac dock must all be reachable. If not, adjust or reroll with seed plus one, up to 5 times, and save the result.

Determinism: every step is driven by a hash of the seed and the chunk coordinates, so the server and all clients get the same world. Chunks are about 64 m and are generated as the player approaches, spread across frames to avoid hitches.

Navigation: from far away, settlements read as silhouettes (water tower, smoke, a chapel spire, lit windows at night), and roads and signs lead to them.

28. Shelters and bases

Shelter types

- **Temporary:** `tent`, `lean_to`, a `sleeping_bag` spot.
- **Found:** `cabin`, `motel_room`, `garage`, `bus_base`, `container_fort`, `lighthouse_outpost`, `bunker`, `treehouse`, `silo`, `school_gym`, `water_tower_base`.
- **Built:** the Bro Basecamp from section 1, now buildable anywhere.

Base systems

- **Claim:** place a `claim_flag` (a cartoon pennant) to claim a building or area. The claim radius grows with base tier: 10, 20 or 35 m. Zombies do not spawn inside a claim. A claim lapses if every owner is away for 5 days.
- **Safe ring:** zombies that are chasing you can still enter, so defenses matter. Sleeping in a bed inside a claim is safe and skips the night.
- **Defenses:** wooden and metal doors, boarded windows, barricades, fences and gates, spike pits, sandbags, searchlights, `foam_turret`, `alarm_cans`.
- **Utilities:** `generator` (burns fuel, powers lights, fridge, workbench, radio and turrets), `rain_barrel`, `solar_panel` (R), `stove`, `fridge` (stops spoilage), `bed`, `workbench`, `bubblebro_cauldron`, `drying_rack`, `chest` (shared), `garage_bay` (car repair with a

lift), `garden_bed`, `chicken_coop` (eggs, and a rooster alarm), `radio_station` (calls traders and events).

- **Trade-off:** generators are loud and lights draw zombies. Power makes you strong and visible.
- **Siege events:** some nights a horde targets a claimed base and attacks its walls. Holding it gives a reward crate. A siren warns at dusk.
- **Base tiers:** 0 camp, 1 shack, 2 fortified house, 3 fort, 4 compound. Each tier needs materials and unlocks more slots and defenses.
- **Co-op:** shared chests with permissions, roles, and ownership that transfers if the owner leaves. Up to 3 claims per party.
- **Goofy decor:** inflatable flamingos, a disco ball, a whoopee cushion trap on the couch, neon signs, gnome guards, and a `trophy_shelf` that shows off your finds.
- **FX:** hammer and nail sounds, boards snapping into place, a lights-on click, a generator rumble, a flag flap, dust when walls take hits.

29. Survivors and traders

- **Existing:** `old_stuffins`, `traveling_salesbro`.
- **Settlement NPCs:** `grumpy_shopkeeper` (barters for shiny_buttons), `jumpy_mechanic` (sells car parts and gives repair quests), `nervous_nurse` (heals and cures infection for a price), `retired_general_of_nothing` (gives terrible military advice and real quests), `hermit_hank` (the hermit cabin trader), `radio_dj_midnight` (a voice on the car radio announcing horde nights and supply drops), `lost_tourist` (an escort quest).
- **Rescue and settle:** rescued survivors can move into your base and take a role: guard, scavenger, cook or mechanic. They need food and a bed.
- **Presentation:** gibberish blip voices with subtitles, a trade window, and a tiny goofy animation per NPC.

30. Grab bag of extra ideas

- **Stealth and movement:** sneak, sprint with stamina, vault fences, cut fences with bolt_cutters, climb ladders to rooftops, ziplines.
- **Weather:** rain muffles noise, fog cuts vision both ways, snow leaves footprints that zombies follow. Floods, landslides and fallen bridges close roads and force detours.
- **Zombie day and night:** slower by day, faster at night, drawn to light and noise.
- **Radio:** the DJ announces horde nights and `supply_drop` crates falling by goofy parachute.

- **Fuel economy:** stations run dry, so siphoning and finding a `gas_can` matter. Tune it so you are never stranded with no way forward.
- **Perks as hats:** use a skill to earn it (lockpicking, mechanic, quiet steps, bargaining) and gain a matching hat.
- **Achievements:** Honk At 100 Zombies, Park A School Bus In A Tree, Survive Horde Night In A Bathtub, Find All The Gnomes.
- **Lore:** notes, graffiti and radio logs telling a silly outbreak story (original, no brand names).
- **Replay:** daily challenge seeds, a speedrun timer, seeded races, and modifiers such as one car only, no ranged weapons, or slapstick mode.
- **Photo mode** with stickers and a postcard export.
- **Pets:** the `pocket_sloth` and a dog that sniffs out loot.
- **Accessibility:** auto-drive assist, colourblind-safe loot tier colours, a reduced-motion option.

31. Plan additions

New phases for the build prompt, in order: items and weapons data, map layout with settlements and roads, settlement and building generation, loot placement, then zombies, vehicles and repair, then shelters and bases, then rosters, combat and NPCs, the quest, multiplayer and polish. Keep each as its own approval step. A progress log at docs/PROGRESS.md records each finished phase so a new session can resume.

32. Part 4: Controls and input layouts

Part 4 defines how the game is controlled on keyboard and mouse, gamepad and phone. Existing controls are kept: WASD, Q and E for the camera, Space for the action, Enter for chat, G for dance, H for honk. New controls fill the gaps.

Principles

- One set of game actions, many input providers. Game logic reads actions (move, action, interact), never raw keys or touches.
- Context-aware: the same control does the right thing for the situation (on foot, driving, repairing, building), and the screen shows only the buttons that matter right now.
- Nothing existing moves. New bindings only fill gaps.
- Thumb-friendly: on a phone, everything used in a fight is reachable without moving your thumbs.

- Prompts match the device: key caps on desktop, icons on phone, button glyphs on gamepad.

33. Keyboard and mouse

On foot

- **W A S D or arrow keys:** move, relative to the camera.
- **Shift (hold):** sprint. Uses stamina and makes more noise.
- **Ctrl (hold):** sneak. Slower and quieter.
- **Q / E:** rotate the camera (existing). Middle-mouse drag also orbits, the mouse wheel zooms, Home recenters behind you.
- **Space or left click:** action: chop, attack, smack (hold to keep going, existing).
- **F:** interact: pick up, open, loot, enter or exit a car, claim, talk. Hold F for long actions such as looting a crate, repairing or claiming.
- **R:** quick use. Eats or heals with the best fitting item in one press.
- **T or right click:** throw the selected throwable. Right click aims, release throws.
- **1 to 8:** hotbar. **[and]:** previous and next slot.
- **Tab:** backpack (inventory, crafting and status tabs). **C:** crafting directly (the existing recipe drawer). **M:** map. **B:** build and base mode.
- **V:** ping a spot for teammates (middle click also works).
- **G:** dance (existing). Hold G for the emote wheel. **H:** honk (existing). **Enter:** chat bubble (existing).
- **Backspace:** drop the selected item.
- **Esc:** pause, or close the open panel.
- **F1:** help. **F2:** photo mode. **Backtick:** debug panel (dev builds only).

Driving

- **W / S:** gas, and brake or reverse. **A / D:** steer.
- **Space:** handbrake and drift. **Shift:** boost, if a booster is fitted.
- **H:** honk (hold for a long honk).
- **X:** start or stop the engine. Hold X to crank and release in the green zone.
- **L:** headlights. **R:** next radio station, hold R for off.
- **F:** exit the car. **1 to 4:** switch seats.
- **C:** car status (parts and fuel). **Tab:** trunk.
- **Q / E:** orbit the camera. **Home:** lock the camera behind the car.

Passenger and gunner

- Mouse to aim, Space or left click to fire or swing from the window, T to throw.
- 1 to 4 to switch seats, F to exit, V to ping.

Repair panel (parked, near the car)

- Hold F near the car to open it. Click a part card to select it. Hold left mouse, Space or F to repair while the ring fills. 1 to 8 picks a tool. Esc closes.
- Other players' name tags show which part they are working on.

Build mode (B)

- Left click places. Right click or Esc cancels. R rotates the piece. Mouse wheel or [and] picks the next piece. Hold Shift for free placement without grid snap.

Inventory

- Drag and drop. Right click for a menu (use, drop, split). Shift-click moves items quickly between backpack and trunk or chest.

34. Gamepad (optional)

- **Left stick:** move. **Right stick:** camera. Click left stick toggles sprint. Click right stick toggles sneak.
- **RT:** action. **LT:** aim and throw. **A:** interact. **X:** quick use. **Y:** backpack. **B:** back or cancel.
- **LB / RB:** previous and next hotbar slot.
- **D-pad:** up for the map, hold down for the emote wheel, left for honk, right for ping.
- **Start:** pause. **Select:** map and objectives.
- **Driving:** RT gas, LT brake and reverse, left stick steer, RB handbrake, Y exit, X engine, A honk, D-pad up headlights, D-pad right radio.
- Aim assist and larger prompts on gamepad.

35. Phone layout (landscape first)

The screen is split into zones by thumb reach. Every touch target is at least 48 dp and spaced at least 12 dp from its neighbours. Everything respects the safe areas around notches and rounded corners.

On foot (default)

- **Bottom-left, floating thumbstick:** it appears wherever the thumb lands in the left 40% of the screen. 12% dead zone, eased direction. Push past 85% of the radius to sprint. A small turtle button above it toggles sneak.

- **Bottom-right, a big ACTION button:** hold to chop or attack, with a haptic tick on each hit. An arc of smaller buttons surrounds it: INTERACT (a hand that changes to a loot, door, car, wrench or flag icon depending on what is nearby), QUICK USE (eat or heal), and THROW (drag from the button to aim, release to throw).
- **Bottom-center, hotbar of 5 slots:** tap to equip, swipe sideways for more, swipe up to open the full backpack.
- **Top-left:** a round minimap with roads and teammates and a compass. Under it two small buttons: chat or emote wheel (dance and quick messages), and honk.
- **Top-center:** stat bars for hunger, health and sanity. Infection, weight and noise meters appear only when relevant.
- **Top-right:** small icons for pause, map and backpack, with the camera snap buttons (rotate left and right) under them.
- **Camera:** drag anywhere in the free area to orbit, pinch to zoom. The two-finger swipe still works (existing).
- **Ping:** long-press on the world or the minimap.

Driving (switches automatically when you enter a car)

- **Left:** steering. Default is two large steer buttons that ramp smoothly the longer you hold. Options: a virtual wheel, or tilt to steer.
- **Right:** a large GAS pedal above a BRAKE and REVERSE pedal, with a smaller HANDBRAKE button beside them. HORN is a big button above the pedals.
- **Top-right row:** exit, headlights, engine key, seat switch.
- **Top-left:** radio. **Top-center:** speed and fuel gauge plus damage warnings.
- Camera orbit by dragging the free area.

Repair panel

- A bottom sheet of part cards (missing, broken, damaged, fine). Tap a card, then press and hold the big FIX button, which is the ACTION button moved into place. The ring fills with haptic ticks and a success pulse.

Build mode

- A ghost piece you drag to position. Buttons on the right: place, cancel, rotate, next piece.

Inventory and trading

- A bottom sheet. Long-press to drag and drop. Tap an item for a menu (use, equip, drop). Double-tap to equip. A weight bar and a sort button sit on top.

Haptics: light taps on chops, a strong pulse on damage, ticks during the engine crank and repairs, and a short pattern for a Jackpot or a rare drop. Everything can be turned off.

Comfort and accessibility

- Left-handed mode that mirrors the layout.
- A button size slider (80% to 130%) and an opacity slider.
- A layout editor: drag buttons where you want them and save.
- Hold versus toggle options for sprint and sneak.
- Auto-aim assist for melee.
- Keep the screen awake, go fullscreen on the first tap, and ask for landscape when you start driving.
- A portrait fallback with only the stick, ACTION and hotbar, plus a rotate prompt for driving.

36. Input architecture and settings

- **Action map:** define actions such as move, sprint, sneak, action, interact, quick_use, throw, hotbar_n, camera_rotate, honk, dance, chat, backpack, craft, map, build, ping, steer, throttle, brake, handbrake, boost, engine, headlights, exit_vehicle and seat_swap. Contexts: on_foot, driving, passenger, repairing, building, ui and chat.
- **Providers:** keyboard and mouse, gamepad and touch each translate raw input into actions. The game reads actions only.
- **Physical keys:** use event.code (physical position) so WASD works on AZERTY and other layouts, and show the correct key label in prompts.
- **Chat safety:** when the chat box is focused, ignore game keys.
- **Browser hygiene:** prevent default on Tab, Space, arrow scrolling and the F-keys the game uses. On touch, use pointer events with touch-action none, track every pointer ID, block double-tap zoom and pull-to-refresh, and release all inputs when the tab loses focus.
- **Buffering:** a 120 ms input buffer so action presses are never eaten.
- **Rebinding:** a settings screen with conflict detection and presets (Default, Arrow keys, Left-handed, One-handed), saved in localStorage on your own site.
- **Prompts:** switch automatically between key caps, touch icons and gamepad glyphs depending on the last device used.
- **Tests:** Playwright keyboard tests for each context, simulated multi-touch tests, and a checklist for real phones (small, large, notched).

37. Plan additions for controls

A new phase Z0B (input system and on-foot layouts) goes right after Z0. Each later phase adds its own context: driving in Z6, repair in Z7, building in Z8, combat and throwing in Z10. Left-handed mode, the layout editor, gamepad support and real-device tests go in Z13.

38. Part 5: Vehicle fleet expansion

Part 5 grows the fleet beyond section 14. Same IDs, tiers (J, C, U, R, X) and quirk rules. Every vehicle is original, goofy, and uses the repair system from section 15.

Vehicle stat model (one VehicleDef per ID): top speed, acceleration, handling, grip, toughness (HP plus part wear), fuel capacity and burn, noise (0 to 5), seats, cargo slots, weight class, height class, offroad rating, water rating and one special ability. Classes: scooter, compact, sedan, van, pickup, truck, bus, heavy, weird, boat, rail, air.

Fleet roles. Every vehicle is good at one job and bad at others, so players choose per trip.

- **Scout:** fast and fragile (`pizza_scooter` , `dune_buggy` , `whirlybird`).
- **Silent:** quiet for sneaking (`golf_cart_deluxe` , `bicycle_bell` , `rowboat_oars` , `scout_balloon`).
- **Hauler:** lots of cargo, slow (`box_truck_moving` , `semi_rig` , `school_bus_of_chaos`).
- **Base on wheels:** sleep and cook on the move (`camper_van_of_comfort` , `food_truck_fiesta` , `party_bus`).
- **Utility:** opens routes (`tow_truck_with_attitude` , `bulldozer_bro` , `fire_truck_of_fun`).
- **Fighter:** smashes zombies (`monster_truck_of_questionable_taste` , `garbage_truck_grumble` , `party_tank`).
- **Terrain specialist:** crosses what others cannot (`swamp_hovercraft` , `snowmobile_zippy` , `quad_bike_atv` , `jet_ski_zoom`).

39. More cars, trucks and big rigs

Cars

- `taxi_of_lies` (C): 4 seats, 6 slots. Rescued survivors who ride pay fares in shiny_buttons. The meter ding counts as a noise event.
- `tiny_bubble_car` (C): 2 seats, 3 slots, very low fuel use. Egg-shaped, tips onto its side on hard turns, and you push it back upright.
- `dune_buggy` (U): 2 seats, fast, the best grip on sand and dirt. No roof, so zombies can reach the seats.

- `vintage_roadster` (U): 2 seats, convertible. Sanity slowly recovers while driving with the top down, with wind-blown hair particles.
- `party_hearse` (U): 3 seats, 10 coffin slots. Spooky organ radio, and Hollow creatures ignore you.
- `patrol_wagon` (U): 4 seats, push bar and siren. The siren turns zombies within 40 m toward you, so use it as a lure.
- `ambulance_pal` (U): 3 seats, a stretcher slot that slowly heals riders, and a siren.
- `stretch_limo` (R): 8 seats, 12 slots, a huge turning circle, a mini fridge that stops food spoiling, and the longest honk.
- `camper_van_of_comfort` (R): 4 seats, 14 slots, a bed, a stove and a fold-out tent. A mobile base with a thirsty engine.
- `hot_tub_hauler` (R): 4 seats, slow. Restores sanity and warmth while you sit in it. Bubble trail, jet-spray horn.
- `couch_on_wheels` (J): 3 seats, mower engine, very slow. Search the cushions daily for 3 shiny_buttons.
- `porta_potty_dragster` (R): 1 seat, rocket speed in a straight line, terrible steering. Leaves a swirl of blue puffs when boosting.

Trucks and big rigs

- `pickup_with_bed` (C): 3 seats, a 12-slot open bed. Two bed riders can swing weapons or throw. Cargo spills on big jumps.
- `box_truck_moving` (U): 3 seats, 40 slots. A zombie sometimes hides in the back when it spawns.
- `food_truck_fiesta` (U): 3 seats, 16 slots, griddle and fryer with more recipes than the hot_dog_van. Toss food through the serving window to teammates. The smell lures zombies and the goose.
- `mail_truck_mailbox` (C): 2 seats, 18 slots. Collect letters from roadside mailboxes as you drive by, for hints.
- `tow_truck_with_attitude` (U): 2 seats. The winch pulls wrecks off roads and tows broken cars.
- `fire_truck_of_fun` (R): 6 seats. The ladder reaches rooftops, and the hose sprays zombies away and puts out fires. Loud siren.
- `garbage_truck_grumble` (U): 3 seats, slow. The compactor crunches zombies into green puffs. Its smell draws zombies and the goose.
- `cement_mixer_maracas` (U): the drum rattles loudly. Drops a gunk puddle behind it that slows zombies.

- `tanker_truck_gassy` (R): 2 seats, carries 500 fuel. Parked at a base it works as a fuel station. If damaged it leaks, with a cartoon fireball risk.
- `armored_bank_truck` (R): 3 seats, very tough, slow. The vault in the back needs a keycard_green and holds jackpot loot.
- `semi_rig` (R): 2 seats. Hitches trailers (section 44) and cannot cross weak bridges.
- `party_bus` (R): 10 seats with disco lights. Disco mode stuns zombies within 15 m for 6 s, but the music is noise level 5 and calls distant hordes.
- `bulldozer_bro` (R): 1 seat, very slow. The blade pushes wrecks and barricades aside. Cannot use boardwalks.
- `road_roller_stan` (R): 1 seat. Flattens zombies into cartoon pancakes and paves dirt tracks into roads as it goes, so other cars drive faster on them.
- `forklift_fred` (U): 1 seat. Lifts crates up to rooftops and nudges roadblocks.
- `party_tank` (X): a boss reward. A confetti cannon turret, crushes cars, extremely tough.

40. Bikes, scooters and tiny rides

- `bicycle_bell` (J): no fuel, silent, 1 seat. The bell is a tiny decoy.
- `bmx_trick_bike` (C): jumps ramps and curbs. Landing a flip gives a small sanity boost.
- `kick_scooter_electric` (C): battery powered and silent. Folds into your backpack.
- `quad_bike_atv` (C): 2 seats, strong off-road.
- `chopper_motorbike` (U): fast, 2 seats, wheelies and leans into turns.
- `sidecar_motorbike` (U): 3 seats. The sidecar rider throws or aims.
- `mobility_scooter_turbo` (U): slow but heavy. Bumps zombies aside with a cheerful beep. Has a flag and a cup holder.
- `unicycle_circus` (R): needs a balance mini-game. Hops over fences and juggles throwables.

41. Boats, snow and swamp rides

Zombies cannot swim, so deep water is a natural escape route. Boats make that route usable.

- `rowboat_oars` (J): silent water crossing.
- `inflatable_dinghy` (C): pool-noodle oars. It can pop.
- `rubber_duck_pedalboat` (U): a giant duck pedal boat that quacks.
- `jet_ski_zoom` (U): fast, 2 seats, loud, short fuel range.

- `speedboat_splash` (U): 4 seats.
- `pontoon_party_boat` (R): 8 seats, slow, with a fire pit. A floating base.
- `fishing_trawler` (R): large. Nets bring in food, and it is one evac option. Needs the boat_key.
- `swamp_hovercraft` (R): crosses mud, water and bog. The fan is loud and the skirt blows dust and ripples.
- `snowmobile_zippy` (U): fast on ice and snow, poor on asphalt.
- `sled_with_sloth` (C): coasts downhill, and a pocket_sloth pulls it at a leisurely pace on flat ground.

42. Rail and rideables

- `rail_handcar` (U): a pump trolley for abandoned rail lines. Two players pump together. Rails are a fast, mostly zombie-free lane, but broken gaps need patching with `track_gap_plates` and switches need flipping. Great co-op.
- `locomotive_of_doom` (X): a late-game train engine on the main rail loop. Needs boiler_fuel and a crew.
- `goose_steed` (R): a tamed honk_goose you ride. Tame it with a `cheese_wheel`. Honk to dash.
- `ostrich_runner` (U, Flats): a fast sprinter that kicks zombies. It panics when you honk.

43. Aircraft: the whirlybird and friends

Build the helicopter first. The rest are optional.

- `whirlybird` (R): a small 2-seat helicopter. Fragile, thirsty and very loud.
- `pedal_copter` (U): human-powered. Pedal to spin the rotor. No fuel, low altitude only.
- `scout_balloon` (U): a hot-air balloon. Silent, drifts with the wind, burns propane. Great for scouting.
- `paraglider_kite` (U): launches from cliffs in Rockbottom Ridge. A silent glide down.
- `crop_duster_biplane` (R): fast. Needs a long straight road as a runway.
- `sky_blimp` (X): an endgame airship base. Very slow.

Whirlybird flight model

- Arcade flight with three altitude bands: ground, low and high. Climb and descend with buttons. The camera zooms out with altitude, and a shadow on the ground shows where you are.

- In the air, zombies cannot reach you. Low over the ground (under about 2 m) they can jump up and grab the skids. Spin the helicopter to fling them off.
- Hover assist is on by default: release the controls and it holds position.
- Landing needs flat, clear ground, a rooftop or a `helipad`. A hard landing damages parts. Land assist lowers you slowly over a safe spot.
- Seats: a pilot and one passenger or winch operator. A `winch_hook` lifts crates, survivors and even a small car on a sling.

Limits, so it does not trivialize the game

- Very loud (noise 5): hordes within a huge radius head toward where you land.
- Burns `aviation_fuel` (rare). Gas works but burns faster and damages the engine.
- Small cargo: only 4 slots, so use the sling for big loads.
- Weather: wind gusts push it, fog grounds it, storms can spin it out, and night flying needs the searchlight.
- Hazards: honk_goose flocks cause bird strikes (feathers everywhere), ash updrafts near the volcano, and engine stalls in Hollow fog.
- Tall objects block low flight: water towers, radio towers, chimneys.

Where to find it and how to unlock it

- One `whirlybird_wreck` spawns per map at a rooftop `helipad` (police_station, hospital_ruins, lighthouse_outpost or volcano_observatory).
- Its parts are scattered across a 3-stop chain: `rotor_blades`, `tail_rotor`, `turbine_engine`, `gearbox`, `landing_skids`, `cockpit_glass` and a goofy `flight_gizmo` that keeps it level.
- Starting needs a keycard and a 4 s rotor spool-up, which is very loud.

Evac alternative. There are three ways out: drive and take the boat from the dock, take the trawler, or fly the whirlybird to the rescue ship. The helicopter is the fastest and riskiest route.

FX: a spinning blade blur, a downwash that flattens grass and kicks up leaves and dust rings, ripples on water, zombies staggering back, a ground shadow that scales with height, a doppler whop-whop with a squeaky rubber-chicken twist, an engine cough when fuel is low, feathers on bird strikes, and a cartoon boing and spin on a crash.

Performance and networking: flying shows far more of the map. Use impostors for distant buildings, never generate interiors from the air, cap airspeed so chunk streaming keeps up, and use a larger interest radius for flying vehicles. The server owns the flight state.

44. Trailers, towing and hitches

- **Hitch:** a `trailer_hitch` part lets a vehicle pull a trailer. Reversing with one is deliberately wobbly and funny.
- **Trailers:** `trailer_cargo` (30 slots), `trailer_camper` (a bed), `trailer_boat` (carry a boat to water), `trailer_flatbed` (carry one small vehicle), `trailer_water_tank`, `trailer_hay_wagon` (soft ride, hides loot).
- **Towing:** a `tow_rope` between two vehicles lets one pull a broken one. The tow truck winch does the same with wrecks.

45. Route rules: weight, height and width

These rules make the fleet matter without touching the biomes. They feed the road generator.

- **Bridge load limits:** light, medium or heavy. A `weak_bridge` collapses under a heavy rig in a cartoon way. Signs show the limit with goofy symbols.
- **Tunnel heights:** tall vehicles (bus, fire truck with the ladder up, semi) cannot enter. Fold the ladder or take a detour.
- **Width:** narrow boardwalks and mountain lanes scrape wide vehicles. Heavy vehicles break boardwalk planks.
- **Tires and runners:** `snow_chains`, `balloon_tires` and `knobby_tires` change grip on snow, mud and sand.
- **Validator:** the reachability test runs per vehicle class (light, medium, heavy, plus each specialist) so a valid route exists for at least three classes, and for the evac dock.

46. Parts and repair for the new vehicles

- **Cars and trucks:** everything in section 15, plus `trailer_hitch`, `winch_cable`, `hydraulic_pump` (ladders and blades) and `siren_box`.
- **Two-wheelers:** `chain`, `pedals`, `handlebars`, `battery_pack`, `kickstand`.
- **Boats:** `hull_patch_kit`, `outboard_motor`, `propeller`, `bilge_pump`, `oars`, `life_jacket`.
- **Snow and swamp:** `sled_runners`, `track_belt`, `fan_blade`, `skirt_fabric`.
- **Helicopter:** `rotor_blades`, `tail_rotor`, `turbine_engine`, `gearbox`, `landing_skids`, `cockpit_glass`, `flight_gizmo`, `aviation_fuel`.
- **Balloon:** `balloon_envelope`, `burner`, `basket`, `propane_tank`.
- **Rail:** `rail_wheels`, `pump_handle`, `track_gap_plates`.

- **New repair mini-games:** rotor balancing (hold and release in the green zone as the blades wobble), hull patching (slap patches on leaks while water rises), chain re-seating (drag the chain back on), and winch rigging.
- **Jank fixes:** `pool_noodle_oars` and `umbrella_sail` for boats, `ceiling_fan_rotor` (works at 40% and wobbles), `trash_bag_balloon`, `bedsheet_wings`.
- **Customization:** paint, decals, bobbleheads, fuzzy dice, underglow neon, exhaust styles (flames, bubbles), horn songs and mounted gear (searchlight, foam turret, plow, ram, winch, roof rack).

47. Controls for the new vehicles

Bikes and scooters: same as driving. Space bunny-hops, Shift pedals faster. The sidecar rider aims with the mouse.

Boats, hovercraft and snowmobiles: same as driving. Space is the brake or anchor instead of drift.

Handcar: alternate A and D to pump (on phone, tap two buttons alternately). Space brakes, F exits.

Whirlybird (pilot), keyboard

- **W / S:** forward and back. **A / D:** turn.
- **Space (hold):** climb. **Ctrl (hold):** descend.
- **Shift:** boost. **Q / E:** orbit the camera (existing).
- **X:** start the engine (4 s rotor spool-up).
- **F:** exit when landed, or hold F while low to rappel.
- **Z:** hold to lower the winch, release to raise it.
- **L:** searchlight. **H:** horn. **R:** radio. **C:** status. **Tab:** cargo. **V:** ping.
- Release everything and it hovers.

Whirlybird, phone

- **Left:** a virtual stick to move, with auto-turn toward your movement.
- **Right:** big Up and Down buttons stacked, with a small Land button that auto-lands over flat ground.
- **Top row:** winch, searchlight, horn, boost and exit or rappel. **Top center:** altitude and fuel gauges.
- Lifting your thumbs makes it hover.

Gamepad: left stick move, right stick turn, RT climb, LT descend, A winch, X searchlight, Y exit, B horn.

Balloon: hold Space for the burner to rise, hold Ctrl to vent and descend. No steering: pick an altitude band to catch a different wind.

48. Where vehicles spawn

- `auto_garage` and houses: cars and the tow truck. `farmhouse`: tractor and pickup. `supermarket`: shopping carts and box trucks.
- `police_station`: patrol_wagon, plus a rooftop helipad. `hospital_ruins`: ambulance_pal and a helipad.
- `harbor_town`: boats and the trawler. `junkyard_town`: weird rides (couch, hot tub, porta potty, road roller).
- `ski_lodge_village`: snowmobiles. `stilt_village`: hovercraft. `desert_trading_post`: dune_buggy and ostrich_runner. `mining_camp`: quad bikes. Ridge cliffs: paraglider launches.
- `abandoned_theme_park`: pedal_copter. Rail depots: the handcar. `volcano_observatory`: a helipad.
- **Guarantees:** each region has at least one common vehicle within reach, and there is exactly one whirlybird wreck per map with its parts chain.

49. Plan additions for the fleet

After the rosters phase (Z9) add: Z9B big rigs, trailers and route rules (including the per-class reachability validator), Z9C bikes, boats, snow, swamp and rail, and Z9D aircraft with the whirlybird first and the others optional. Each is its own approval step.

50. Phone layouts for the whirlybird and boats

These match the mockups on the control design canvas (artboards Phone: whirlybird and Phone: boat). Same rules as section 35: landscape first, targets 48 px or larger, 12 px gaps, safe areas respected, left-handed mode mirrors everything.

Whirlybird (pilot seat). The layout switches on automatically.

- **Bottom-left, floating stick:** appears under the thumb in the left 40% of the screen, 12% dead zone. It moves the helicopter in any direction, and the helicopter turns toward where you push. Lifting the thumb makes it hover (hover assist).
- **Bottom-right, stacked pair:** a big Climb button (about 104 by 108) above a Descend button (about 104 by 86). Hold to climb or descend through the ground, low and high altitude bands.

- **Land:** a small round button (64 px) to the left of the pair. It is only active over flat, clear ground, a rooftop or a helipad, and then a prompt appears. It lowers the helicopter slowly and safely.
- **Top-right row, five buttons:** winch (hold to lower, release to raise), searchlight, horn, boost, and exit or rappel (hold when low to rappel down).
- **Top-center panel:** the altitude band (Ground, Low, High, with the active band highlighted) and a fuel bar.
- **Warning chips under the panel:** zombie on the skids (spin to shake it off), low fuel, wind gust, bird strike.
- **Engine off:** a Start engine button appears in the center. Hold it for the 4 s rotor spool-up. Show a noise ring, because it is loud.
- **Camera:** drag empty space to orbit, pinch to zoom, and zoom out automatically with altitude.
- **Haptics:** a rumble while the rotor spools up, a thump on landing, a click on the winch, and a pulse on warnings.

Boat (and the other water rides)

- **Same layout as the driving layout in section 35**, so thumbs already know it: two steer buttons bottom-left, a big pedal pair bottom-right, a horn above them, and a top-right row of seat switch, engine key, lights and leave boat. Radio top-left, gauges top-center, drag empty space to orbit.
- **Differences:** the big pedal is labeled Throttle. The lower pedal is Reverse (tap to slow down, hold to go backward). Anchor replaces Drift and holds the boat in place. The gauges show speed, fuel and hull. Exit means leave the boat.
- **Leaks:** when the hull takes on water, a warning chip appears and tapping it opens the hull patching mini-game as a bottom sheet.
- **Hovercraft and snowmobile** reuse this layout with Drift in place of Anchor.

Input contexts to add: boat, air (whirlybird), balloon and handcar, in addition to the contexts listed in section 36.

51. Plan additions for the new layouts

Implement the whirlybird layout in the aircraft phase (Z9D) and the boat layout in the boats phase (Z9C). Reuse the driving layout components wherever possible so there is one set of buttons with different labels. Add Playwright touch tests: lift both thumbs and confirm the helicopter hovers, confirm Land is disabled over rough ground, confirm the Start engine hold takes 4 seconds, and confirm the hull patching sheet opens from a leak warning. The layout editor in Z13 must cover every vehicle layout.